

SANTHOSH M

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PROFESSIONAL SUMMARY

AI/ML Engineer specializing in production-grade Generative AI systems, Retrieval-Augmented Generation pipelines, LLM fine-tuning workflows, and intelligent multi-agent architectures. Demonstrated track record of cutting AI hallucination rates, building scalable knowledge retrieval systems, and delivering end-to-end ML solutions across healthcare, sports analytics, and information retrieval domains.

TECHNICAL SKILLS

AI & Machine Learning: LLMs, Generative AI, RAG, Fine-tuning (QLoRA/LoRA), NLP, NER, spaCy, Scikit-learn
Frameworks & Libraries: LangChain, HuggingFace Transformers, PEFT, TRL, PyTorch, Pandas, NumPy, FastAPI
Databases & Vector Stores: Milvus, Neo4j, PostgreSQL, MySQL, ChromaDB
Cloud & DevOps: Docker, HuggingFace Spaces, Render, Supabase, Cloudflare Pages, GitHub Actions
APIs & Search: Tavily API, DuckDuckGo Search, Wikipedia REST API, BeautifulSoup, Requests
Languages & Tools: Python, PHP, SQL, Laravel, Jupyter Notebook, Tableau, ArcGIS 10.5

PROFESSIONAL EXPERIENCE

AI Developer — LezdotechMed

Oct 2024 – Present

Salem, Tamil Nadu, India

- Architected and launched production RAG systems for medical diagnostic support using LangChain, slashing AI hallucination by 40% through semantic vector grounding across curated knowledge bases.
- Spearheaded the development of multi-source retrieval agents combining Tavily API, DuckDuckGo, and BeautifulSoup web scraping, with Claude as the core reasoning engine and a tiered fallback strategy for uninterrupted service.
- Established a Neo4j-backed clinical QA audit pipeline with fault-tolerant transaction management, guaranteeing 100% data integrity across every query-response cycle.

Machine Learning Intern — Giggso

Jul 2024 – Sep 2024

Chennai, Tamil Nadu, India

- Trained and shipped Scikit-learn models covering classification, regression, and clustering tasks; wired inference outputs directly into a user-facing GUI for real-time interaction.
- Leveraged NLP preprocessing techniques — tokenization, stopword removal, and sentiment scoring — on scraped Amazon review data to surface actionable product insights.

PROJECTS

CricBot IPL — AI-Powered IPL Intelligence Chatbot

Solo Developer | 2026

- Fine-tuned TinyLlama-1.1B on 17 years of IPL data (2008–2025) with QLoRA/PEFT and a Milvus Lite vector retrieval pipeline, backed by a Wikipedia REST API fallback for out-of-domain queries.
- Incorporated spaCy NER-driven session memory across 10-turn conversation windows, franchise-adaptive UI theming for all 10 IPL teams, and a fully zero-cost hosting stack.

Tech Stack: PyTorch, QLoRA, HuggingFace PEFT/TRL, TinyLlama-1.1B, Milvus Lite, FastAPI, spaCy, Docker, Supabase, Cloudflare Pages

AI-Powered Medical Diagnostic Assistant

Solo Developer | 2024 | Generative AI / RAG

- Constructed a multi-LLM RAG pipeline for context-aware clinical information retrieval; curbed model hallucination by 40% by anchoring responses to structured medical knowledge stored in ChromaDB.

Tech Stack: LangChain, RAG, ChromaDB, Python, HuggingFace

AI-Powered Multi-Source Information Retrieval Agent

Solo Developer | 2024 | LLM Agent Architecture

- Orchestrated a LangChain agent unifying Tavily API, DuckDuckGo, and a bespoke BeautifulSoup scraper; configured Claude as the central reasoning engine with cascading fallback logic for consistent, high-fidelity responses.

Tech Stack: LangChain, Claude API, Tavily API, DuckDuckGo, BeautifulSoup, Python

Neo4j-Augmented Medical QA System

Solo Developer | 2024 | Graph Database + Generative AI

- Formulated a multi-tiered clinical QA solution drawing on Tavily API, DuckDuckGo, and Claude-2 for generative synthesis; implemented a Neo4j graph audit layer with explicit transaction controls for end-to-end interaction traceability.

Tech Stack: Neo4j, LangChain, Claude-2, Tavily API, Python, FastAPI

Machine Learning Deployment with GUI Integration

Solo Developer | 2024 | ML Engineering

- Assembled a full-spectrum ML deployment suite — spanning classification, regression, and clustering — and integrated it with a Tkinter GUI, enabling real-time inference and interpretability dashboards for non-technical stakeholders.

Tech Stack: Scikit-learn, Python, Pandas, NumPy, Matplotlib, Tkinter

Amazon Review Sentiment Analysis — NLP & Web Scraping

Solo Developer | 2024 | NLP

- Automated large-scale review harvesting from Amazon via BeautifulSoup and Requests; executed NLP analysis — tokenization, stopword removal, and polarity scoring — to extract consumer sentiment trends.

Tech Stack: Python, BeautifulSoup, Requests, NLTK, Pandas, Matplotlib

Demarcation of Groundwater Potential Zones — Geospatial Analysis

Research Project | 2022 | Edappadi Block, Salem District, Tamil Nadu

- Conducted remote sensing and spatial mapping with ArcGIS 10.5 to delineate groundwater potential zones; synthesized geological and hydrological datasets to support regional environmental planning initiatives.

Tech Stack: ArcGIS 10.5, Remote Sensing, Geospatial Analysis, PostgreSQL

EDUCATION

M.Phil in Applied Geology

Government Arts College (Autonomous), Salem

2019 – 2022

M.Sc in Applied Geology

Government Arts College (Autonomous), Salem

2017 – 2019

CERTIFICATIONS

Generative AI Certified Professional — Oracle

Salesforce AI Associate — Salesforce

Nanodegree in Data Science — Imagecon India Private Limited, Salem

LANGUAGES

English (Professional Proficiency) | **Tamil** (Native)